# Numerical inequalities

What "greater than" really means — one definition by subtraction that settles every comparison, however awkward the numbers.

LESSON 38–39 | 2 × 40 MIN | ALGEBRA 8, §11 | STAGE 9 · 4.3

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

| | |
|---|---|
| Warm-up | 5 min |
| Explanation | 12 min |
| Interactive | 8 min |
| Practice | 13 min |
| Homework | 2 min |

LEARNING OBJECTIVES

- State the definition of $a > b$ by means of the difference.
- Compare two numbers or expressions using the difference method.
- Read and write strict and non-strict inequalities.
- Place numbers correctly on a number line.

TEXTBOOK

| | |
|---|---|
| National | §11, pp. 68–70 |
| Cambridge | Learner's Book pp. 96–102 |
| Workbook | Workbook 4.3 |

## 01 Explanation

### The definition

DEFINITION

$$a > b \text{ means } a - b > 0$$

$$a < b \text{ means } a - b < 0$$

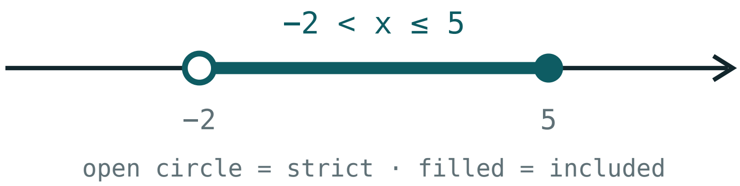
and $a = b$ means $a - b = 0$.

That is the whole of this lesson. Comparing two numbers is not a matter of looking at them — it is a matter of **subtracting and checking the sign**. For ordinary numbers this feels like overkill; for expressions with letters it is the only method that works.

On the number line, $a > b$ means $a$ lies to the **right** of $b$.

### Strict and non-strict

| SYMBOL | READ AS | TYPE | ON THE LINE |
|---|---|---|---|
| $a > b$ | a is greater than b | strict | open circle |
| $a < b$ | a is less than b | strict | open circle |
| $a \geq b$ | a is greater than or equal to b | non-strict | filled circle |
| $a \leq b$ | a is less than or equal to b | non-strict | filled circle |




*An open circle excludes the boundary; a filled circle includes it.*

Note that $5 \geq 5$ is **true** — “or equal to” only has to hold on one side.

## The difference method in action

Compare $\frac{3}{7}$ and $\frac{4}{9}$ without decimals:

$$\frac{3}{7} - \frac{4}{9} = \frac{27 - 28}{63} = -\frac{1}{63} < 0$$

The difference is negative, so $\frac{3}{7} < \frac{4}{9}$. No decimals, no calculator, no rounding to argue about.

With letters it is the only route. To show that $a^2 + 1 > 2a$ for every $a$, subtract:

$$a^2 + 1 - 2a = (a - 1)^2 \geq 0$$

A square is never negative, so the difference is never negative — the inequality holds, with equality only when $a = 1$.

TWO USEFUL FACTS

$a^2 \geq 0$ for every $a$, and $a^2 > 0$ unless $a = 0$. Almost every proof in this chapter ends by spotting a square.

## 02 Worked examples

**EXAMPLE 1** ***Compare $\frac{5}{8}$ and $\frac{7}{11}$***

1. $\frac{5}{8} - \frac{7}{11}$

   Subtract.

2. $= \frac{55-56}{88} = -\frac{1}{88}$

   Common denominator 88.

3. The difference is negative.

4. $\frac{5}{8} < \frac{7}{11}$

ANSWER

$\frac{5}{8} < \frac{7}{11}$

EXAMPLE 2 ***Prove that $a^2 + 4 \geq 4a$ for every $a$***

1. $a^2 + 4 - 4a$

   Take the difference.

2. $= a^2 - 4a + 4$

   Rearrange.

3. $= (a - 2)^2$

   A perfect square.

4. $(a - 2)^2 \geq 0$ always.

   So the difference is never negative.

ANSWER

True for every $a$, with equality at $a = 2$.

EXAMPLE 3 ***Which is greater: $2^{10}$ or $10^3$?***

1. $2^{10} = 1024$

2. $10^3 = 1000$

3. $1024 - 1000 = 24 > 0$

   The difference is positive.

4. $2^{10} > 10^3$

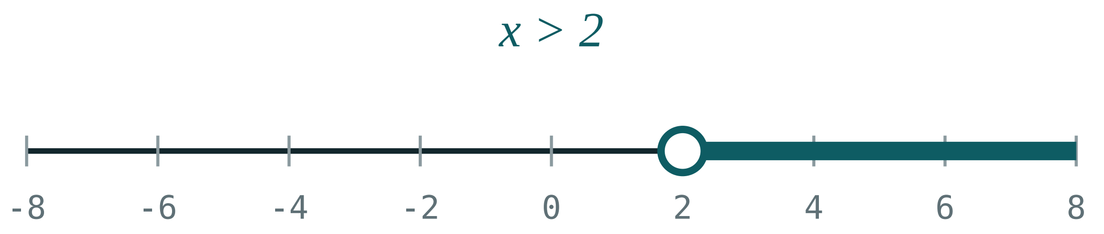
ANSWER

$2^{10} > 10^3$

## 03 Interactive model

Move the boundary and flip the symbol — the open and filled circles are the whole story.

**Solutions on the number line**




inequality `x > 2` | boundary `2 (open)` | x = 0 works? **no** | x = 6 works? **yes**

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| Inequality | Tengsizlik | Неравенство |
| Greater than | Katta | Больше |
| Less than | Kichik | Меньше |
| Strict inequality | Qat’iy tengsizlik | Строгое неравенство |
| Non-strict inequality | Qat’iy bo‘lmagan tengsizlik | Нестрогое неравенство |
| Difference | Ayirma | Разность |
| Positive | Musbat | Положительный |
| Negative | Manfiy | Отрицательный |
| Number line | Sonlar o‘qi | Числовая прямая |
| Compare | Taqqoslash | Сравнить |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

# 05 Quick check

### Check yourself

$a > b$ means:

- $a - b > 0$
- $b - a > 0$
- $a + b > 0$
- $ab > 0$

Which statement is true?

- $5 > 5$
- $5 \geq 5$
- $5 < 5$
- none of them

$a^2$ is:

- always positive
- never negative
- always negative
- sometimes negative

On the number line, a filled circle means:

- the value is excluded
- the value is included
- the inequality is strict
- nothing in particular

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up 7 PROBLEMS

1. *Compare 7 and 4 using the difference*

   ANSWER $7 - 4 = 3 > 0$, so $7 > 4$

2. *Compare −3 and −5*

   ANSWER $-3 - (-5) = 2 > 0$, so $-3 > -5$

3. *Is $6 \geq 6$ true?*

   ANSWER Yes.

4. *Is $6 > 6$ true?*

   ANSWER No.

5. *Write “x is at least 5” as an inequality*

   ANSWER $x \geq 5$

6. *Write “x is less than −2” as an inequality*

   ANSWER $x < -2$

7. *Compare $\frac{1}{2}$ and $\frac{2}{5}$*

   ANSWER $\frac{1}{2} > \frac{2}{5}$

**Medium · the standard question** 7 PROBLEMS

1. *Compare* $\frac{3}{7}$ *and* $\frac{4}{9}$

ANSWER $\frac{3}{7} < \frac{4}{9}$

2. *Compare* $\frac{5}{8}$ *and* $\frac{7}{11}$

ANSWER $\frac{5}{8} < \frac{7}{11}$

3. *Which is greater:* $2^{10}$ *or* $10^3$*?*

ANSWER $2^{10} = 1024 > 1000$

4. *Compare* $3\sqrt{2}$ *and* $4$

ANSWER $3\sqrt{2} \approx 4.24 > 4$

5. *Prove* $a^2 + 1 \geq 2a$

ANSWER The difference is $(a - 1)^2 \geq 0$.

6. *Prove* $a^2 + 4 \geq 4a$

ANSWER The difference is $(a - 2)^2 \geq 0$.

7. *Compare* $-\frac{2}{3}$ *and* $-\frac{3}{4}$

ANSWER $-\frac{2}{3} > -\frac{3}{4}$

## Hard · stretch and reason

7 PROBLEMS

1. *Prove that* $a^2 + b^2 \geq 2ab$ *for all* $a, b$

ANSWER The difference is $(a - b)^2 \geq 0$.

2. *Prove that* $a + \frac{1}{a} \geq 2$ *for* $a > 0$

ANSWER The difference is $\frac{(a-1)^2}{a} \geq 0$ because $a > 0$.

3. *Compare* $\frac{a}{b}$ *and* $\frac{a+1}{b+1}$ *for* $a > b > 0$

ANSWER The difference is $\frac{a-b}{b(b+1)} > 0$, so $\frac{a}{b} > \frac{a+1}{b+1}$.

4. *Prove that* $(a + b)^2 \geq 4ab$

ANSWER The difference is $(a - b)^2 \geq 0$.

5. *Which is greater:* $\sqrt{10}$ *or* $3.2$?

ANSWER $10 < 10.24 = 3.2^2$, so $\sqrt{10} < 3.2$

6. *Show that* $a^2 + b^2 + 2 \geq 2a + 2b$

ANSWER The difference is $(a - 1)^2 + (b - 1)^2 \geq 0$.

7. *For which* $a$ *is* $a^2 > a$?

ANSWER $a^2 - a = a(a - 1) > 0$ when $a < 0$ or $a > 1$.

## 07 Homework

### Homework — 5 problems

Algebra 8, §11, pp. 68–70. Show the difference in every comparison.

1. *Compare* $\frac{4}{9}$ *and* $\frac{5}{11}$ *by the difference method*

2. *Compare* $-\frac{3}{5}$ *and* $-\frac{5}{8}$
3. *Prove that* $a^2 + 9 \geq 6a$ *for every* $a$
4. *Which is greater:* $3^5$ *or* $5^3$*?*
5. *Write as inequalities: “x is at most 7” and “y is greater than −1”*